

Linear Inequality:

$$1) -2(m-3) < 5(m+1) - 12$$

$$-2m + 6 < 5m + 5 - 12$$

$$-2m + 6 < 5m - 7$$

$$6 + 7 < 5m + 2m$$

$$13 < 7m$$

$$7m > 13$$

$$m > \frac{13}{7}$$

$$= \left(\frac{13}{7}, \infty \right)$$

$$2) 2(1-x) + 5 \leq 3(2x-1)$$

$$2 - 2x + 5 \leq 6x - 3$$

$$7 - 2x \leq 6x - 3$$

$$7 + 3 \leq 6x + 2x$$

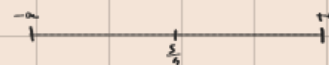
$$10 \leq 8x$$

$$8x \geq 10$$

$$x \geq \frac{10}{8}$$

$$x \geq \frac{5}{4}$$

$$\left[\frac{5}{4}, \infty \right)$$



$$3) -14 < -7(3x+2) < 1$$

$$-14 < -7(3x+2)$$

$$-7(3x+2) < 1$$

$$-14 < -21x - 14$$

$$-21x - 14 < 1$$

$$-21x - 14 > -14$$

$$-21x < 1 + 14$$

$$-21x > 0$$

$$-21x < 15$$

$$x < 0$$

$$x > -\frac{15}{21}$$

$$-\frac{15}{21} < x < 0$$

$$= \left(-\frac{15}{21}, 0 \right)$$

$$4) -3 < \frac{3}{2}(2-x) \leq 5$$

$$-3 < \frac{3}{2}(2-x)$$

$$\frac{3}{2}(2-x) \leq 5$$

$$-3 < 3 - \frac{3}{2}x$$

$$3 - \frac{3}{2}x \leq 5$$

$$-3 - 3 < -\frac{3}{2}x$$

$$-\frac{3}{2}x \leq 5 - 3$$

$$-\frac{3}{2}x < +6$$

$$-\frac{3}{2}x \leq 2$$

$$3x < 12$$

$$-3x \leq 4$$

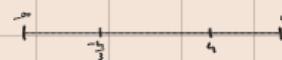
$$x < \frac{12}{3}$$

$$x \geq -\frac{4}{3}$$

$$x < 4$$

$$-\frac{4}{3} \leq x < 4$$

$$= \left[-\frac{4}{3}, 4 \right)$$



Quadratic Inequalities:

1) $2x^2 - 7x + 3 > 0$

$$2x^2 - (6+1)x + 3 > 0$$

$$2x^2 - 6x - x + 3 > 0$$

$$2x(x-3) - 1(x-3) > 0$$

$$(2x-1)(x-3) > 0$$

$$2x-1=0 \quad x-3=0$$

$$x = \frac{1}{2} \quad x = 3$$



$$x \in (-\infty, \frac{1}{2}) \cup (3, \infty)$$

2) $-x^2 + 6x + 7 \geq 0$

$$-x^2 + (-1+7)x + 7 \geq 0$$

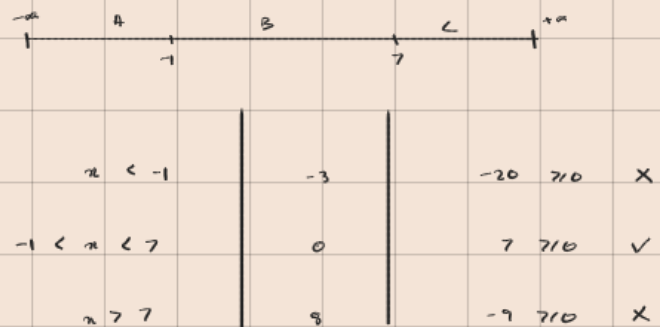
$$-x^2 - x + 7x + 7 \geq 0$$

$$-x(x+1) + 7(x+1) \geq 0$$

$$(7-x)(x+1) \geq 0$$

$$7-x=0 \quad x+1=0$$

$$x = 7 \quad x = -1$$



$$x \in [-1, 7]$$

3) $x^2 - 2x - 11 \leq 0$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-11)}}{2(1)}$$

$$x = \frac{+2 \pm \sqrt{4 + 44}}{-2}$$

$$x = -1 - 2\sqrt{3} \quad x = -1 + 2\sqrt{3}$$

$$4) \quad x^2 > x + 2$$

$$x^2 - x - 2 > 0$$

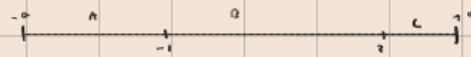
$$x^2 - (2-1)x - 2 > 0$$

$$x^2 - 2x + x - 2 > 0$$

$$x(x-2) + 1(x-2) > 0$$

$$(x-2)(x+1) > 0$$

$$x = 2 \quad x = -1$$



$$x < -1$$

$$-2$$

$$4 > 0 \quad \checkmark$$

$$-1 < x < 2$$

$$0$$

$$-2 > 0 \quad \times$$

$$x > 2$$

$$3$$

$$9 > 0 \quad \checkmark$$

$$x \in (-\infty, -1) \cup (2, \infty)$$

Rational Inequalities:

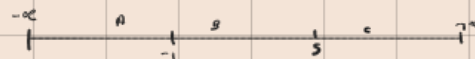
$$1) \quad \frac{x+1}{x-5} \leq 0$$

$$x+1 = 0$$

$$x-5 = 0$$

$$x = -1$$

$$x = 5$$



$$x < -1$$

$$-2$$

$$\frac{1}{2} \leq 0 \quad \times$$

$$-1 < x < 5$$

$$0$$

$$-\frac{1}{5} \leq 0 \quad \checkmark$$

$$x > 5$$

$$6$$

$$7 \leq 0 \quad \times$$

$$x \in [-1, 5)$$

$$2) \quad \frac{x^2 + 4x + 3}{x-1} > 0$$

$$x-1$$

$$x^2 + (3+1)x + 3 > 0$$

$$x-1$$

$$x^2 + 3x + x + 3 > 0$$

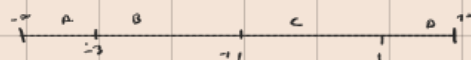
$$x-1$$

$$x(x+3) + 1(x+3) > 0$$

$$x-1$$

$$\frac{(x+1)(x+3)}{(x-1)} > 0$$

$$x = 1, \quad x = -1, \quad x = -3$$



$$x < -3$$

$$-4$$

$$-\frac{3}{2} > 0 \quad \checkmark$$

$$-3 < x < -1$$

$$-2$$

$$\frac{1}{3} > 0 \quad \checkmark$$

$$-1 < x < 1$$

$$0$$

$$-5 > 0 \quad \times$$

$$x > 1$$

$$2$$

$$15 > 0 \quad \checkmark$$

$$x \in (-3, -1) \cup (1, \infty)$$

$$3) \quad \underline{x^2 - 16} < 0$$

$$(x-1)^2$$

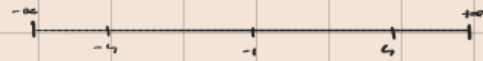
$$\underline{x^2 - 4^2} < 0$$

$$(x-1)(x-1)$$

$$\underline{(x-4)(x+4)}$$

$$(x-1)(x-1)$$

$$x=4, \quad x=-4, \quad x=1$$



$$x < -4$$

$$-5$$

$$\frac{1}{4} < 0 \quad \times$$

$$-4 < x < -1$$

$$-2$$

$$-\frac{2}{3} < 0 \quad \checkmark$$

$$x > 4$$

$$5$$

$$\frac{5}{16} < 0 \quad \times$$

$$x \in (-4, -1)$$